identical to those in the inflationary picture, so the resulting nonuniformities predicted by the two theories are nearly identical as well.[11] Moreover, while there isn't an inflationary burst in the cyclic model, there is a trillion-year period (beginning at Stage 3) of milder accelerated expansion. But it's really just a matter of haste versus patience; what the inflationary model accomplishes in a flash, the cyclic model accomplishes in a comparative eternity. Since the collision in the cyclic model is not the beginning of the universe, there is the luxury of slowly resolving cosmological issues (like the flatness and horizon problems) during the final trillion years of each *previous* cycle. Eons of gentle but steady accelerated expansion at the end of each cycle stretch our three-brane nice and flat, and, except for tiny but important quantum fluctuations, make it thoroughly uniform. And so the long, final stage of each cycle, followed by the splat at the beginning of the next cycle, yields an environment very close to that produced by the short surge of expansion in the inflationary approach.

A Brief Assessment

At their present levels of development, both the inflationary and the cyclic models provide insightful cosmological frameworks, but neither offers a complete theory. Ignorance of the prevailing conditions during the universe's earliest moments forces proponents of inflationary cosmology to simply assume, without theoretical justification, that the conditions required for initiating inflation arose. If they did, the theory resolves numerous cosmological conundrums and launches time's arrow. But such successes hinge on inflation's happening in the first place. What's more, inflationary cosmology has not been seamlessly embedded within string theory and so is not yet part of a consistent merger of quantum mechanics and general relativity.

The cyclic model has its own share of shortcomings. As with Tolman's model, consideration of entropy buildup (and also of quantum mechanics[12]) ensures that the cyclic model's cycles could not have gone on forever. Instead, the cycles began at some definite time in the past, and so, as with inflation, we need an explanation of how the first cycle got started. If it did, then the theory, also like inflation, resolves the key cosmological problems and sets time's arrow pointing from each low-